

Quantum Network-Based Prediction of Cancer Driver Genes

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(Dated: August 17, 2025)

Identification of cancer driver genes is fundamental for the development of targeted therapeutic interventions. The integration of mutational profiles with protein-protein interaction (PPI) networks offers a promising avenue for their detection [1, 2], but scaling to large networks is computationally demanding. Quantum computing offers compact representations and potential complexity reductions, and while it has been applied to other network-based biological inference tasks, including link prediction, it has not yet been used for driver gene prediction. We present a supervised quantum framework that combines mutation scores with network topology via a novel state preparation scheme, *Quantum Multi-order Moment Embedding* (QMME), motivated by the classical method of Gumpinger et al. [3]. QMME encodes low-order statistical moments over the mutation scores of a node’s immediate and second-order neighbors, and encodes this information into quantum states. These are used as inputs to a kernel-based quantum binary classifier that discriminates known driver genes from others. Simulations on a real PPI network demonstrate competitive performance relative to classical baselines. A brief complexity analysis suggests the approach could achieve a quantum speedup in network-based cancer gene prediction. This work underscores the potential of supervised quantum graph learning frameworks to advance biological discovery.

I. INTRODUCTION

A central objective in oncology is to identify and catalogue genes underlying human cancer [4–6]. While genes with high mutation rates are likely drivers, most *cancer driver genes* have few mutations. In these cases, frequency-based methods alone cannot reliably distinguish rare cancer driver genes from irrelevant ones [5]. One explanation lies in the networked nature of cellular processes: genes operate within pathways and complexes, so cancer may result from disruption at the pathway level rather than mutations in specific individual genes. Recent research has embraced this interaction-based perspective of cancer, integrating biological networks with statistical measures of gene-cancer associations to identify novel candidate cancer driver genes [1, 7–9]. In these networks, nodes correspond to genes and edges represent relationships between them. Protein-protein interaction (PPI) networks [10–12] are a widely used representation of gene interactions, as they integrate information from diverse data sources, tissues, and molecular processes.

Previous studies have combined PPI networks with gene-level cancer association scores, using clustering or network propagation methods. A common choice for these gene scores is the MutSig *P*-value [13, 14], which estimates the significance of observed mutations by considering (i) the overall mutational burden, (ii) clustering of mutations within a gene’s coding sequence, and (iii) the functional impact of mutations. NetSig, for instance, focuses on the local neighborhood of each gene to identify cancer driver genes while correcting for node degree bias [1]. A few studies have explored supervised approaches, incorporating existing knowledge of known driver genes during

prediction. Gumpinger et al. used Cancer Gene Census (CGC) annotations to train classifiers for discovering new cancer driver genes [3]. This method models the task as a node-level supervised classification problem in a molecular interaction network, combining network topology with mutation-score features. It achieved more than a 10% improvement in area under the precision–recall curve (AUPRC) over baseline methods.

Network-based approaches derive their strength from integrating weak mutation signals across interconnected genes rather than relying on strong evidence from individual genes. This demonstrates the value of graph-structured representations as a paradigm for cancer driver gene discovery. However, as biological networks continue to grow in size and complexity, classical methods face increasing challenges in scalability and representational efficiency. This motivates the exploration of quantum computing as an alternative framework for representing and processing such structured data, including PPI networks. In particular, quantum machine learning (QML), has been proposed as a tool with potential for inference tasks [15–18]. Recent work has begun applying quantum computing to biological network inference problems, including link prediction tasks [19–21], demonstrating potential speedups over classical baselines, further supporting the potential of quantum approaches in this domain.

Classical methods often rely on local statistical descriptors, such as neighborhood means or low-order moments, to capture structural information in graphs [3, 22]. However, quantum embedding strategies that incorporate such statistics remain unexplored. Moreover, many existing QML approaches neglect the gate complexity of state preparation, limiting their practicality as end-to-end

quantum solutions [23, 24]. Motivated by these gaps, we introduce a quantum protocol that embeds low-order statistical moments of features in the one-hop and two-hop neighborhoods into amplitude-encoded quantum states.

In this work, we introduce the *Quantum Multi-order Moment Embedding* (QMME) protocol, which encodes local graph statistics into quantum states for inference. We apply this feature map to a PPI network, where nodes represent genes and features correspond to MutSig P -values. Thus, we design an almost fully quantum pipeline for node-level binary classification of genes as cancer drivers. Classification is performed using the swap-test classifier [25], a non-parametric, distance-based method that employs a quantum kernel based on state similarity.

The entire process, from embeddings to inference, is carried out within the quantum domain, requiring no classical optimization. It relies only on minimal data access assumptions, enabling straightforward circuit design, and involves only limited classical post-processing, inherent to the swap-test classifier. The pipeline explicitly addresses state preparation with practical circuit construction, ensuring embeddings can be constructed with logarithmic qubit resources and constant-depth circuits. We validate our approach through numerical simulations on real graphs, evaluating classification performance alongside quantum resource metrics such as circuit depth and query complexity. This work establishes a proof-of-principle for end-to-end quantum learning on graph-structured biological data and lays groundwork for future quantum models in complex network inference tasks.

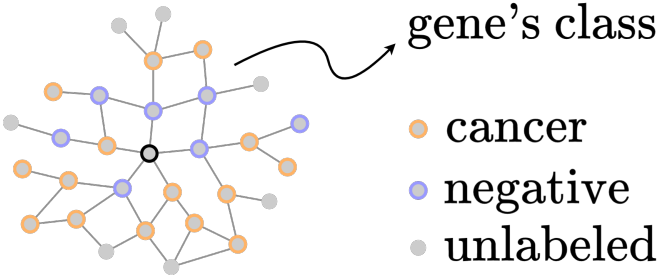


FIG. 1. **Prediction of cancer driver genes.** A PPI network where nodes (genes) are colored by class. The aim is to classify genes and identify cancer drivers based on network structure.

II. PROBLEM SETUP

A. Notation

Protein-protein interaction (PPI) networks model interactions between genes as nodes connected by edges as in fig. 1. Each gene's association with cancer can be quantified using the MutSig P -value [13, 14], which reflects the statistical significance of mutations considering mutational burden, clustering, and functional impact [3].

Formally, we represent a PPI network as an unweighted, undirected graph $\mathcal{G} = (V, E, \mathbf{x})$ with $|V| = N$ nodes, where $V = [N]$ denotes the set of node indices and $E \subseteq V \times V$ is the edge set. The connectivity of $\mathcal{G}$ is described by its adjacency matrix $A \in \{0, 1\}^{N \times N}$, where $A_{vu} = 1$ indicates the presence of an edge between nodes v and u , and $A_{vu} = 0$ its absence. Define the immediate neighborhood of v as

$$\mathcal{N}_v^1 = \{u \in V \mid A_{vu} = 1\}. \quad (1)$$

We assume $\mathcal{N}_v^1$ is ordered by increasing node index. This allows us to define the function $r(v, \ell)$ as

$$r(v, \ell) := \mathcal{N}_v^1(\ell), \quad \ell \in \{0, \dots, |\mathcal{N}_v^1| - 1\}, \quad (2)$$

which returns the index of the ℓ -th neighbor of v in $\mathcal{N}_v^1$. Equivalently, $r(v, \ell)$ is the column index of the ℓ -th nonzero entry in row v of A . If $\ell \geq |\mathcal{N}_v^1|$, i.e., the function returns a flag value indicating no such neighbor exists. The two-hop neighborhood $\mathcal{N}_v^2 \subseteq V$ is defined as

$$\mathcal{N}_v^2 = \{w \in V \mid \exists u \in V, A_{vu} = 1 \wedge A_{uw} = 1\}. \quad (3)$$

Each node $v \in V$ has a scalar feature value $x_v \in \mathbb{R}$, representing its MutSig P -value quantifying the gene's mutation significance. The collection of all node features is denoted by the vector

$$\mathbf{x} = (x_1, x_2, \dots, x_N) \in \mathbb{R}^N. \quad (4)$$

B. Node Embeddings

The low-order origin (or raw) moments of a scalar random variable $X \sim \mathcal{P}$ are defined as $\mathbb{E}[X^i]$, where i denotes the order of the moment [26]. These moments are uncentered (i.e., calculated about the origin) and capture fundamental properties of the distribution. The first four are the mean $\mathbb{E}[X]$, the uncentered quadratic $\mathbb{E}[X^2]$, cubic $\mathbb{E}[X^3]$, and quartic $\mathbb{E}[X^4]$ moments. Low-order origin moments serve as the building blocks for the more familiar summary statistics of variance, skewness, and kurtosis, which are centered or standardized variants derived from them. Formally, we define the *origin moment embedding* as the mapping

$$\bar{\varphi}(X) = [\mathbb{E}[X], \mathbb{E}[X^2], \mathbb{E}[X^3], \mathbb{E}[X^4]], \quad (5)$$

which collects the first four origin moments of X . Since the true distribution $\mathcal{P}$ is generally unknown, these moments are typically estimated empirically. For a finite sample $\mathbf{s} = [s_1, \dots, s_k]$ of X , with $s_j \in \mathbb{R}$, the corresponding *sample raw moment embedding* is given by

$$\varphi(\mathbf{s}) = \left[\sum_{j=1}^k \frac{s_j}{k}, \sum_{j=1}^k \frac{s_j^2}{k}, \sum_{j=1}^k \frac{s_j^3}{k}, \sum_{j=1}^k \frac{s_j^4}{k} \right]. \quad (6)$$

which returns the first four sample origin moments from the sample, thus, describing its distributional shape. The hop order $c \in \{1, 2\}$ specifies the neighborhood distance from node v . Given $\mathbf{x}$ defined in eq. (4),

$$\mathcal{X}_v^c = \{x_u \in \mathbf{x} \mid u \in \mathcal{N}_v^c\}, \quad (7)$$

denotes the set of feature values of nodes at hop-distance c from v , which are samples from a local distribution $\mathcal{P}_v^c$. The function $\varphi(\cdot)$ is applied to each $\mathcal{X}_v^c$, such that

$$\mathbf{m}_v^c := \varphi(\mathcal{X}_v^c) \in \mathbb{R}^4. \quad (8)$$

For vertex $v \in V$, the node embedding is then the concatenation of $\mathbf{m}_v^c$ from immediate and two-hop neighborhoods:

$$\mathbf{m}_v = [\mathbf{m}_v^1, \mathbf{m}_v^2] \in \mathbb{R}^8, \quad (9)$$

defining the *node embedding function* $\mathbf{m}_v : V \rightarrow \mathbb{R}^8$.

C. Problem Statement

We consider a supervised binary classification task on a graph $\mathcal{G} = (V, E, \mathbf{x})$, where a subset of nodes $V_l \subseteq V$ is labeled. For these nodes, $y_v = 1$ for $v \in V_1 \subset V_l$ and $y_v = 0$ for $v \in V_0 \subset V_l$, corresponding to the positive and negative classes, respectively. This is a node-level classification task, i.e., the aim is to predict labels for individual nodes. Our goal is to predict the label $\hat{y}_v$ for any node $v \in V$, using a method that combines a *quantum node embedding* unitary operator $U_{\mathcal{G}}$, derived from the node features $\mathbf{x}$ and the connectivity E of $\mathcal{G}$, with a quantum binary classifier $\mathcal{C}$. This combination is used to predict whether a node v belongs to the positive or negative class. That is,

$$\mathcal{C} \left(U_{\mathcal{G}} |v\rangle |0\rangle^{\otimes \mathbf{m}} \right) = \hat{y}_v, \quad (10)$$

where $\hat{y}_v \in \{0, 1\}$ denotes the predicted class label for node v , and $\mathbf{m}$ is the number of data qubits.

The objective is to explicitly define $U_{\mathcal{G}}$ as the state preparation step and design the full quantum circuit for the supervised classification of cancer driver genes. The input is a PPI network where the node features $\mathbf{x}$ are per-gene MutSig P -values. For the labeled data, a set of positively labeled genes V_1 is obtained from CGC data, so the positive class corresponds to cancer driver genes.

III. QUANTUM MULTI-ORDER MOMENT EMBEDDING (QMME)

A. Input Model

We base our algorithm on having query access to the graph $\mathcal{G} = (V, E, \mathbf{x})$ through the adjacency list model for

bounded-degree graphs [27–29]. In this input model, the graph has N vertices and bounded degree $s = 2^s$, i.e., at most s nonzero entries in each row of the adjacency matrix [30, 31]. To accommodate an attributed network, we extend this model by adding a feature query. Thus, the three available query operations are the following:

1. **neighbor query:** given $v \in V$ and $\ell \in \mathbb{Z}$, returns the ℓ -th neighbour of v if $\ell \leq k_v$, and $*$ otherwise,
2. **feature query:** given $v \in V$, returns the value x_v ,
3. **degree query:** given $v \in V$, returns the degree k_v .

A quantum equivalent of this input model can be described by defining three corresponding unitary operators O_L , $\tilde{O}_X$, and $\tilde{O}_K$, acting on appropriate Hilbert spaces. Here, O_L provides access to a node's immediate neighborhood, $\tilde{O}_X$ encodes node features, and $\tilde{O}_K$ allows access to the number of c -hop neighbors. The so-called *oracles* act as:

$$O_L : |v\rangle |\ell\rangle \mapsto |v\rangle |r(v, \ell)\rangle, \quad (11)$$

$$\tilde{O}_X : |v\rangle |0^{d'}\rangle \mapsto |v\rangle |\tilde{x}_v\rangle, \quad (12)$$

$$\tilde{O}_K : |v, c\rangle |0^{d'}\rangle \mapsto |v, c\rangle |\tilde{k}_v^c\rangle, \quad (13)$$

where v is the node index, ℓ the neighbor index, $\tilde{x}_v$ the d' -bit fixed point representation of the node feature value $x_v \in \mathbf{x}$, and $\tilde{k}_v^c$ the binary representation of the node degree $k_v^c = |\mathcal{N}_v^c|$, i.e., the number of c -hop neighbors.

Although oracle implementation on quantum hardware remains an active topic of research [32], our work assumes coherent access, allowing queries to be made in superposition, which is a standard assumption in the theoretical framework of quantum algorithms. We also assume access to the controlled versions of the oracles O_L , $\tilde{O}_X$, and $\tilde{O}_K$ and their inverses. Since these oracles are unitary and represented by real-valued matrices, their inverses reduce to their transposes. Neighbor and feature queries, whether classical or quantum, are counted as $\mathcal{O}(1)$ in query complexity. The degree query can be emulated by $\mathcal{O}(\log s)$ neighbor queries via binary search [28]. We introduce $\tilde{O}_K$ as a generalization of the degree query, enabling access to the size of a c -hop neighborhood, with $c \in \{1, 2\}$. Thus, $\tilde{O}_K$ can be constructed from O_L with $\mathcal{O}(\log^2 s)$ queries.

B. Rotation Operators

The goal of the embedding circuit is to encode local graph statistics into the amplitudes of quantum states. This is done using controlled single-qubit rotations, where the rotation angles are determined by values encoded in the computational basis and retrieved from the oracles. To this end, we define two rotation operators, F_X and F_{K-1} . Given a node feature $x_v \in \mathbb{R}$ with $|x_v| \leq 1$, and its binary encoding $\tilde{x}_v$, the *feature rotation operator* F_X [30]

is defined as

$$F_X : |0\rangle |\tilde{x}_v\rangle \mapsto \left(x_v |0\rangle + \sqrt{1 - |x_v|^2} |1\rangle \right) |\tilde{x}_v\rangle, \quad (14)$$

which corresponds to a single-qubit rotation $R_y(\theta)$ with angle $\theta = 2 \arcsin(x_v)$. This encodes the scalar value x_v into the amplitude of the $|0\rangle$ component. Similarly, we define the *degree-inverse rotation operator* $F_{K^{-1}}$ [33] as

$$F_{K^{-1}} : |0\rangle |\tilde{k}_v^c\rangle \mapsto \left(\frac{1}{k_v^c} |0\rangle + \sqrt{1 - \left| \frac{1}{k_v^c} \right|^2} |1\rangle \right) |\tilde{k}_v^c\rangle. \quad (15)$$

These operators are implemented using multi-controlled R_y rotation gates [30], where the control register holds the binary input $\tilde{x}_v$ or $\tilde{k}_v^c$. The rotation angles are $\theta(\tilde{x}_v) = 2 \arcsin(x_v)$ and $\theta(\tilde{k}_v^c) = 2 \arcsin(1/k_v^c)$, respectively.

C. Feature and Degree-Inverse Oracles

To amplitude-encode the node embedding $\mathbf{m}_v$ into quantum states, we introduce two derived oracles O_X and $O_{K^{-1}}$, which use the basis-encoding input oracles $\tilde{O}_X$ and $\tilde{O}_{K^{-1}}$, along with rotation operators F_X and $F_{K^{-1}}$. The *feature oracle* O_X encodes powers of the feature value x_v conditioned on the control register $|i\rangle$. Specifically, the amplitude of the component where the ancilla register is in state $|0\rangle^{\otimes 4}$ is mapped to x_v^i , which relates to the i -th order term in the raw moment embedding function $\varphi(\cdot)$ from eq. (6). The oracle O_X acts on a control register, where the integer i in the control register $|i\rangle$ dictates how many of the ancilla qubits are rotated by F_X , as follows:

$$\begin{aligned} |v, i\rangle |0\rangle^{\otimes (4+d')} &\xrightarrow{\tilde{O}_X} |v, i\rangle |0\rangle |\tilde{x}_v\rangle \\ &\xrightarrow{C-F_X} |v, i\rangle \left(x_v |0\rangle + \sqrt{1 - |x_v|^2} |1\rangle \right)^{\otimes i} |0\rangle^{\otimes (4-i)} |\tilde{x}_v\rangle \\ &\xrightarrow{\tilde{O}_X^T} |v, i\rangle \left(x_v |0\rangle + \sqrt{1 - |x_v|^2} |1\rangle \right)^{\otimes i} |0\rangle^{\otimes (4-i+d')}. \end{aligned} \quad (16)$$

Omitting the intermediate register $|\tilde{x}_v\rangle$, fig. 2 summarizes the circuit of oracle O_X , which can be expressed as

$$O_X : |v, i\rangle |0\rangle^{\otimes 4} \mapsto |v, i\rangle \left(x_v |0\rangle + \sqrt{1 - |x_v|^2} |1\rangle \right)^{\otimes i} |0\rangle^{\otimes (4-i)}. \quad (17)$$

The degree-inverse oracle $O_{K^{-1}}$ amplitude-encodes the inverse degree scaling factors $1/k_v^c$, required to emulate the average present in the moment embedding function $\varphi(\cdot)$, included in the node embedding $\mathbf{m}_v$. It performs a controlled rotation that encodes the value $1/k_v^c$ as the

amplitude of the $|0\rangle$ state of an ancilla qubit, as follows:

$$\begin{aligned} |v, c\rangle |0\rangle^{\otimes (1+d')} &\xrightarrow{\tilde{O}_K} |v, c\rangle |0\rangle |\tilde{k}_v\rangle \\ &\xrightarrow{F_{K^{-1}}} |v, c\rangle \left(\frac{1}{k_v^c} |0\rangle + \sqrt{1 - \left| \frac{1}{k_v^c} \right|^2} |1\rangle \right) |\tilde{k}_v\rangle \\ &\xrightarrow{\tilde{O}_K^T} |v, c\rangle \left(\frac{1}{k_v^c} |0\rangle + \sqrt{1 - \left| \frac{1}{k_v^c} \right|^2} |1\rangle \right) |0\rangle^{\otimes d'}. \end{aligned} \quad (18)$$

As shown in fig. 3, the intermediate register $|\tilde{k}_v\rangle$ is omitted, and the oracle is represented as:

$$O_{K^{-1}} : |v, c\rangle |0\rangle \mapsto |v, c\rangle \left(\frac{1}{k_v^c} |0\rangle + \sqrt{1 - \left| \frac{1}{k_v^c} \right|^2} |1\rangle \right). \quad (19)$$

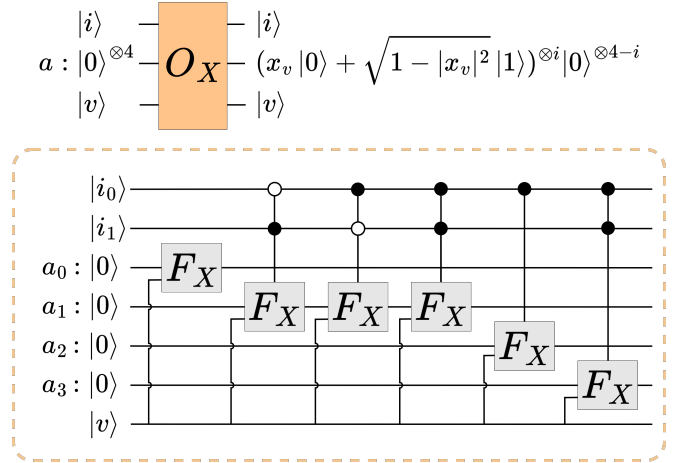


FIG. 2. **Circuit of the feature oracle O_X , based on controlled versions of the rotation operator F_X .** A two-qubit control register $|i\rangle$ selects how many of the four ancilla qubits undergo the rotation F_X : for $i = 00$, only the first qubit; for $i = 01$, the first two; for $i = 10$, the first three; and for $i = 11$, all four. A d' -qubit register is used internally but omitted for clarity.

Together with the neighbor oracle O_L and Hadamard gates, the oracles O_X and $O_{K^{-1}}$ form the core building blocks of the QMME circuit, as depicted in Figure 3.

D. QMME Algorithm

We define the QMME unitary operator U_G as

$$U_G : |v\rangle |0\rangle^{\otimes \mathbf{m}} \mapsto |v\rangle |\psi_v\rangle, \quad (20)$$

and refer to this mapping as the *quantum node embedding*. The second register, consisting of $\mathbf{m}$ qubits, is called the *data register*. The total number of qubits is

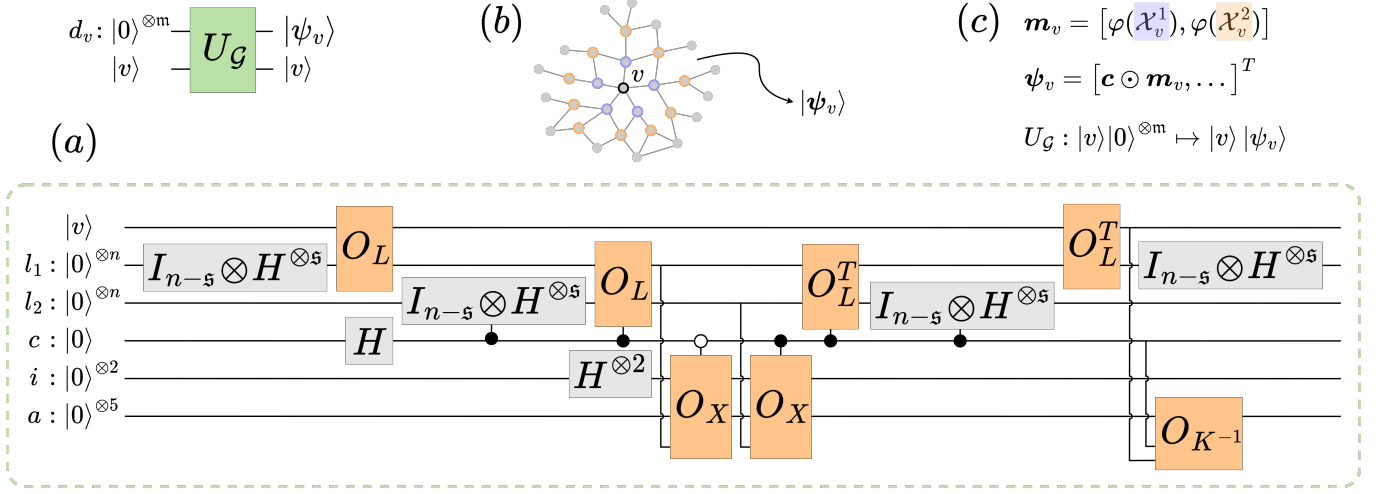


FIG. 3. **Quantum Multi-order Moment Embedding (QMME) protocol.** (a) Circuit-level implementation of the unitary operator $U_G : |v\rangle |0\rangle^{\otimes m} \mapsto |v\rangle |\psi_v\rangle$, which amplitude-encodes low-order statistical moments of node v 's neighborhood, shown in (b). Here, $|v\rangle$ encodes the node index $v \in V$, and d_v is the m -qubit data register initialized to zero with $m = \mathcal{O}(\log N)$, composed of registers for immediate and second-hop neighbors (ℓ_1, ℓ_2), a control qubit c , a two-qubit register i indexing the i -th moment, and a five-qubit ancilla register a . The circuit accesses the network in (b) via the quantum oracles in eq. (13). (b) Input PPI network $\mathcal{G} = (V, E, \mathbf{x})$ with node features $\mathbf{x}$ representing mutation scores. Immediate neighbors of v are shown in purple, second-hop in orange. These provide the data for the embedding $|\psi_v\rangle \in \mathbb{R}^{2^m}$. (c) The node embedding function $\mathbf{m}_v$, based on the moments $\varphi(\cdot)$ over v 's neighborhood, is encoded as the quantum state $|\psi_v\rangle$.

$m = 2 \log N + 8$ (excluding work registers), and generally scales as $\mathcal{O}(\log N)$. A schematic of the operator U_G circuit is shown in Figure 3, and step-by-step derivation of the quantum node embedding in eq. (20) is presented in the Appendix. The resulting state $|\psi_v\rangle$ consists of two components:

$$|\psi_v\rangle = |\psi_m\rangle + |\psi_\perp\rangle, \quad (21)$$

where $|\psi_m\rangle$ is the *neighborhood-moment component*, capturing the quantum analogue of the classical embedding $\mathbf{m}_v \in \mathbb{R}^8$, and $|\psi_\perp\rangle$ is an orthogonal component.

According to the circuit in Figure 3, $|\psi_m\rangle$ is defined as

$$|\psi_m\rangle = |\mathbf{c} \odot \mathbf{m}_v\rangle |0\rangle^{\otimes (m-3)} \quad (22)$$

$$= \sum_{c \in [2]} \frac{1}{2^{3/2} s^c} \sum_{i \in [4]} \varphi_i(\mathcal{X}_v^c) |c, i\rangle |0\rangle^{\otimes (m-3)} \quad (23)$$

$$= \sum_{j \in [8]} c_j \cdot m_{v,j} |j\rangle |0\rangle^{\otimes (m-3)}, \quad (24)$$

where $\odot$ denotes the element-wise (Hadamard) product, $\varphi_i(\mathcal{X}_v^c)$ are the i th-order moment of the neighborhood $\mathcal{X}_v^c$, and the constant vector $\mathbf{c} \in \mathbb{R}^8$ is

$$\mathbf{c} = \frac{1}{2^{3/2}} \left[\frac{1}{s}, \frac{1}{s}, \frac{1}{s}, \frac{1}{s}, \frac{1}{s^2}, \frac{1}{s^2}, \frac{1}{s^2}, \frac{1}{s^2} \right]. \quad (25)$$

The orthogonal component $|\psi_\perp\rangle$ of the resulting state is

$$|\psi_\perp\rangle = \sum_{a \neq 0} |\psi_{\perp,a}\rangle |a\rangle, \quad (26)$$

where $|a\rangle$ denotes computational basis states in the $(m-3)$ data qubits, and $|\psi_{\perp,a}\rangle \in \mathbb{R}^{2^{(m-3)}}$.

As an example, take the mean of the mutation scores of the immediate neighbors of gene v , i.e., $m_{v,1} = \varphi_1(\mathcal{X}_v^1)$. As in eq. (23), the amplitude of the state $|0_c, 0_i, 0^{(m-3)}\rangle$ will be $\frac{\varphi_1(\mathcal{X}_v^1)}{2^{3/2} s}$. As shown in fig. 3, this arises by:

1. preparing an equal superposition over register ℓ_1 ;
2. O_L retrieves the immediate neighbors' indices;
3. operator O_X encodes the neighbors' feature values into the $|0\rangle$ component of the first ancilla qubit;
4. uncomputing the basis register ℓ_1 via O_L^T leaves the $|0\rangle^{\otimes n}$ component amplitude proportional to the sum of neighbors' features, while others form interference patterns with alternating signs;
5. finally, the ancilla rotation O_{K-1} normalizes by the degree k_v^c , resulting in the mean of $\mathcal{X}_v^c$ encoded in the amplitude of the rotated ancillas at zero.

This procedure describes the action of the unitary U_G for $c, i = 0$. It generalizes to $c \in [2], i \in [4]$, covering immediate and second neighbors, and more moments, yielding

$$\langle v, c, i | \langle 0 |^{\otimes (m-3)} U_G |v\rangle |0\rangle^{\otimes m} = \frac{1}{2^{3/2} s^c} \sum_{x_u \in \mathcal{X}_v^c} \frac{x_u^i}{k_v^c} = \frac{\varphi_i(\mathcal{X}_v^c)}{2^{3/2} s^c}. \quad (27)$$

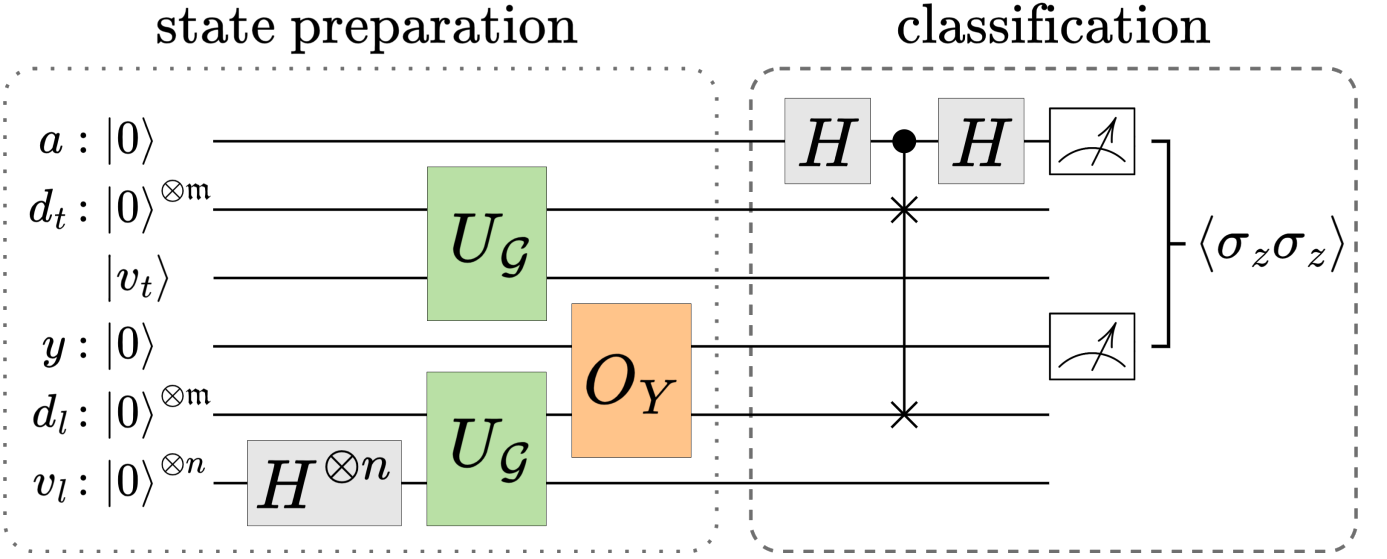


FIG. 4. **Full cancer gene prediction pipeline with the QMME protocol as part of state preparation.** The first register corresponds to the ancilla qubit a , the second to the test data qubits d_t , the third to the index qubits of the test node v_t , the fourth stores the label qubit y , the fifth the labeled data qubits d_l , and the sixth the index qubits of the labeled nodes v_l . The operator U_G , together with the Hadamard gate and the oracle O_Y , prepares the quantum state, which is then processed by the classifier $\mathcal{C}$, i.e., the swap-test classifier. Node labels $\hat{y}_v \in \{0, 1\}$ are predicted via a swap-test and two-qubit measurement statistics on the ancilla and label registers. An outcome of label one indicates that the test gene is predicted as a cancer driver.

IV. QUANTUM CLASSIFICATION FRAMEWORK

The quantum node embedding operator U_G maps each node in a feature-labeled network to its statistical moments, amplitude-encoded within a quantum state $|\psi_v\rangle$. Given a protein-protein interaction (PPI) network as input, the QMME protocol encodes local statistical moments of the MutSig P -values. Depicted in Figure 4, the quantum classification framework assumes query access to the class labels $y_{v_l} \in \{0, 1\}$ for each labeled node $v_l \in V_l$:

1. **label query:** given $v_l \in V_l$, returns the label y_{v_l} ,

with a quantum equivalent defined by the operator O_Y :

$$O_Y : |v_l\rangle |0\rangle \mapsto |v_l\rangle |y_{v_l}\rangle, \quad (28)$$

encoding the class label $y_{v_l} \in \{0, 1\}$ in a single qubit. The proposed quantum classification framework proceeds as follows: after preparing the equal superposition over labeled nodes, the embedding unitary U_G is applied both to this register v_l and to the test node register v_t . The class labels are loaded onto the label qubit via O_Y . Classification follows via the swap-test classifier [25], where the measurement statistics determine the final class prediction of the test node. The full pipeline may be described

as a sequence of quantum state evolutions:

$$\begin{aligned} & |0\rangle |0\rangle^{\otimes m} |v_t\rangle |0\rangle |0\rangle^{\otimes m} |0\rangle^{\otimes n} \\ & \xrightarrow{H_{v_l}^{\otimes n}} \frac{1}{\sqrt{|V_l|}} \sum_{v_l \in V_l} |0\rangle |0\rangle^{\otimes m} |v_t\rangle |0\rangle |0\rangle^{\otimes m} |v_l\rangle \\ & \xrightarrow{U_G} \frac{1}{\sqrt{|V_l|}} \sum_{v_l \in V_l} |0\rangle |\psi_t\rangle |v_t\rangle |0\rangle |\psi_l\rangle |v_l\rangle \\ & \xrightarrow{O_Y} \frac{1}{\sqrt{|V_l|}} \sum_{v_l \in V_l} |0\rangle |\psi_t\rangle |v_t\rangle |y_l\rangle |\psi_l\rangle |v_l\rangle \\ & \xrightarrow{H_a \cdot \text{c-swap} \cdot H_a} |\psi_f\rangle = \frac{1}{\sqrt{|V_l|}} \sum_{v_l \in V_l} (|0\rangle |\psi_+\rangle + |1\rangle |\psi_-\rangle) |v_l\rangle, \end{aligned} \quad (29)$$

where $|\psi_{\pm}\rangle = |\psi_t\rangle (|v_t\rangle |y_l\rangle) |\psi_l\rangle \pm |\psi_l\rangle (|v_t\rangle |y_l\rangle) |\psi_t\rangle$. The swap-test classifier computes the fidelity, i.e., the squared overlap between two states. This yields $|\langle\psi|\phi\rangle|^2$ for two pure states $|\psi\rangle$ and $|\phi\rangle$, and more generally $\text{Tr}(\rho\sigma)$ for two possibly mixed states ρ and σ . Fidelity serves as a similarity measure (one if the states are identical, zero if they are orthogonal), making it a natural distance measure in the quantum feature space. Classification thus relies directly on distances between training and test data, unifying feature mapping and classification into a single quantum procedure without requiring an intermediate classical support vector machine.

The measurement and post-processing involves evaluating the expectation value of the two-qubit observable $\sigma_z^{(a)} \sigma_z^{(l)}$, where the superscripts indicate the ancilla and

label qubit. For the final state $|\psi_f\rangle$, $\langle\sigma_z^{(a)}\sigma_z^{(l)}\rangle$ is given by

$$\langle\sigma_z^{(a)}\sigma_z^{(l)}\rangle = \text{Tr}\left(\sigma_z^{(a)}\sigma_z^{(l)}|\psi_f\rangle\langle\psi_f|\right) \quad (30)$$

$$= \frac{1}{|V_l|} \sum_{v_l \in V_l} (-1)^{y_{v_l}} |\langle\psi_t|\psi_{v_l}\rangle|^2. \quad (31)$$

In practice, these statistics are obtained by repeating r times the two-qubit projective measurement in the σ_z basis. Using $\text{Tr}(\sigma_z|y_{v_l}\rangle\langle y_{v_l}|) = 1$ for $y_{v_l} = 0$ and -1 for $y_{v_l} = 1$, the expectation value is calculated as

$$\langle\sigma_z^{(a)}\sigma_z^{(l)}\rangle = \frac{1}{r} (c_{00} - c_{01} - c_{10} + c_{11}), \quad (32)$$

where c_{al} counts measurements with the ancilla in state a and the label qubit in state l . The test data are classified as 0 if $\langle\sigma_z^{(a)}\sigma_z^{(l)}\rangle > 0$ and 1 if negative, i.e.,

$$\hat{y}_v = \frac{1}{2} \left(1 - \text{sgn} \langle\sigma_z^{(a)}\sigma_z^{(l)}\rangle\right) \quad (33)$$

$$= \mathcal{C} \left(U_{\mathcal{G}} |v\rangle |0\rangle^{\otimes m} \right). \quad (34)$$

which resembles the earlier formulation in Equation (10). Here, $\mathcal{C}$ is the swap-test classifier and $U_{\mathcal{G}}$ corresponds to the Quantum Multi-order Moment Embedding (QMME), encoding local properties in the network $\mathcal{G}$.

V. RESULTS AND ANALYSIS

The implementation and all data used in this work are openly available on GitHub.

A. Dataset Description

We construct our dataset from the OmniPath protein-protein interaction (PPI) network, accessed via the `OmniPathR` R package, which captures pairwise gene interactions at the protein level. Each node is annotated with its $-\log_{10}$ MutSig P -value [14], and nodes without MutSig values are removed. The resulting network contains 6 850 genes and 27 075 interactions, forming the candidate set for node classification. Ground-truth labels are obtained from the Cancer Gene Census (CGC) in the COSMIC database [34]. Tier 1 genes have documented cancer-relevant activity and mutations that promote oncogenic transformation, while Tier 2 genes show strong but less extensive evidence. Both Tier 1 and Tier 2 CGC genes are treated as positives, totaling 590 cancer genes ($\sim 8.6\%$ of the dataset). The remaining genes are considered negative for the purposes of this work, though they are not confirmed negatives and may include undiscovered cancer drivers.

B. Performance Evaluation

To address class imbalance in supervised classification, we balance the labeled set by randomly selecting an equal number of negatives from the unlabeled pool, which is particularly important when performing superposition over labeled nodes.

C. Complexity Analysis

We analyze the complexity of the embedding circuit $U_{\mathcal{G}}$ in Table I. Let d' denote the precision (in qubits) used to encode both feature values and node degrees. The number of oracle calls required for a single application of $U_{\mathcal{G}}$ is constant. The Hadamard gates contribute a cost that scales as $\mathcal{O}(\log s)$ within the unitary operator $U_{\mathcal{G}}$, and as $\mathcal{O}(\log |V_l|)$ across the entire quantum pipeline. Overall, the dominant contribution to the total gate complexity arises from arithmetic operations that are part of oracle subroutines (see the Appendix for a detailed breakdown).

TABLE I. **Complexity analysis for the QMME protocol.** $U_{\mathcal{G}}$ complexity estimates for n -sparse and general graphs, in terms of gate count, circuit depth, qubit requirements, and query complexity, i.e., calls to the input model.

Metric	Adjacency Matrix	
	s -sparse	General
Gate complexity	$\mathcal{O}(\text{polylog} N + d')$	$\mathcal{O}(\text{polylog} N + d')$
Circuit depth	$\mathcal{O}(1)$	$\mathcal{O}(1)$
Qubit count	$\mathcal{O}(\text{polylog} N + d')$	$\mathcal{O}(\text{polylog} N + d')$
Query complexity	$\mathcal{O}(\log^2 s)$	$\mathcal{O}(\log^2 N)$

Circuit depth is the number of sequential layers of gates, i.e., the computation length, and measures time complexity. The number of qubits is the size of the quantum register, which measures space (memory) complexity. Thus, computation time is independent of network size, and space complexity is logarithmic on it.

VII. DISCUSSION AND CONCLUSIONS

In this work, we introduce an almost fully quantum pipeline for cancer driver gene prediction based on a swap-test classifier, requiring no classical optimization loops. All stages of encoding, embedding, and inference are performed within the quantum domain, requiring only minimal data access assumptions and limited classical post-processing. Our approach demonstrates logarithmic resource scaling together with solid performance on protein-protein interaction (PPI) network data. The proposed *Quantum Multi-order Moment Embedding* (QMME) is a general and extensible architecture for encoding local

graph statistics into quantum states. While we focus on binary classification of genes as cancer drivers using PPI networks (with nodes representing genes and MutSig P -values as features), the QMME framework is broadly applicable to node-level classification tasks on feature-labeled graphs across domains [22]. Future directions include extending QMME to capture higher-order moments, supporting weighted graphs, enabling multi-class classification, and implementing resource-efficient variants suitable for near-term quantum hardware.

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Quantum Node Embedding Step-by-step Derivation

U_r denotes the operator U acting on register r , with identity on all other registers; $C - U_r$ the controlled application of U_r , with the control qubit in register c being in state $|1\rangle$; and $C^{(0)} - U_r$ indicates control on the $|0\rangle$ state of c .

$|v, 0^{\mathbf{m}}\rangle$ where $\mathbf{m} = 2 \log N + 8$, omitting work qubits,

$$\begin{aligned}
& \xrightarrow{I_{n-s} \otimes H_{t_1}^{\otimes s}} \frac{1}{\sqrt{s}} |v, 0^3\rangle \sum_{\ell_1 \in [s]} | \ell_1, 0^{(n+5)} \rangle \\
& \xrightarrow{(O_L)_{v, \ell_1}} \frac{1}{\sqrt{s}} |v, 0^3\rangle \sum_{\ell_1 \in [s]} |r(v, \ell_1), 0^{(n+5)}\rangle \\
& \xrightarrow{H_c} \frac{1}{\sqrt{2s}} |v\rangle \sum_{c \in [2]} \sum_{\ell_1 \in [s]} |c, 0^2, r(v, \ell_1), 0^{(n+5)}\rangle \\
& \xrightarrow{I_{n-s} \otimes C-H_{\ell_2}^{\otimes s}} \frac{1}{\sqrt{2s}} |v\rangle \sum_{\ell_1 \in [s]} \left(|0_c, 0^2, r(v, \ell_1), 0^{(n+5)}\rangle + \frac{1}{\sqrt{s}} \sum_{\ell_2 \in [s]} |1_c, 0^2, r(v, \ell_1), \ell_2, 0^5\rangle \right) \\
& \xrightarrow{C-(O_L)_{\ell_1, \ell_2}} \frac{1}{\sqrt{2s}} |v\rangle \sum_{\ell_1 \in [s]} \left(|0_c, 0^2, r(v, \ell_1), 0^{(n+5)}\rangle + \frac{1}{\sqrt{s}} \sum_{\ell_2 \in [s]} |1_c, 0^2, r(v, \ell_1), r(\mathcal{N}_v^1(\ell_1), \ell_2), 0^5\rangle \right) \\
& \xrightarrow{H_i^{\otimes 2}} \frac{1}{2^{3/2} \sqrt{s}} |v\rangle \sum_{\ell_1 \in [s]} \sum_{i \in [4]} \left(|0_c, i, r(v, \ell_1), 0^{(n+5)}\rangle + \frac{1}{\sqrt{s}} \sum_{\ell_2 \in [s]} |1_c, i, r(v, \ell_1), r(\mathcal{N}_v^1(\ell_1), \ell_2), 0^5\rangle \right) \\
& \xrightarrow{C^{(0)}-(O_X)_{i, a, t_1}} \frac{1}{2^{3/2} \sqrt{s}} |v\rangle \sum_{\ell_1, i} \left((x_{r(v, \ell_1)}^i |0_c, i, r(v, \ell_1), 0^{(n+4)}\rangle + |\perp\rangle) |0\rangle + \frac{1}{\sqrt{s}} \sum_{\ell_2 \in [s]} |1_c, i, r(v, \ell_1), r(\mathcal{N}_v^1(\ell_1), \ell_2), 0^5\rangle \right) \\
& \xrightarrow{C-(O_X)_{i, a, \ell_2}} \frac{|v\rangle}{2^{3/2} \sqrt{s}} \sum_{\ell_1, i} \left(x_{r(v, \ell_1)}^i |0_c, i, r(v, \ell_1), 0^{(n+5)}\rangle + \frac{1}{\sqrt{s}} \sum_{\ell_2 \in [s]} x_{r(\mathcal{N}_v^1(\ell_1), \ell_2)}^i |1_c, i, r(v, \ell_1), r(\mathcal{N}_v^1(\ell_1), \ell_2), 0^5\rangle \right) + |v\rangle |\perp\rangle |0\rangle \\
& \xrightarrow{C-(O_L^T)_{\ell_1, \ell_2}} \frac{1}{2^{3/2} \sqrt{s}} |v\rangle \sum_{\ell_1, i} \left(x_{r(v, \ell_1)}^i |0_c, i, r(v, \ell_1), 0^{(n+5)}\rangle + \frac{1}{\sqrt{s}} \sum_{\ell_2 \in [s]} x_{r(\mathcal{N}_v^1(\ell_1), \ell_2)}^i |1_c, i, r(v, \ell_1), \ell_2, 0^5\rangle \right) + |v\rangle |\perp\rangle |0\rangle \\
& \xrightarrow{I_{n-s} \otimes C-H_{\ell_2}^{\otimes s}} \frac{1}{2^{3/2} \sqrt{s}} |v\rangle \sum_{\ell_1, i} \left(x_{r(v, \ell_1)}^i |0_c, i, r(v, \ell_1)\rangle + \frac{1}{\sqrt{s}} \sum_{x' \in \mathcal{X}_{r(v, \ell_1)}^1} x'^i |1_c, i, r(v, \ell_1)\rangle \right) |0^{(n+5)}\rangle + |v\rangle |\perp\rangle |0\rangle \\
& \xrightarrow{(O_L^T)_{v, t_1}} \frac{1}{2^{3/2} \sqrt{s}} |v\rangle \sum_{\ell_1, i} \left(x_{r(v, \ell_1)}^i |0_c, i, \ell_1\rangle + \frac{1}{s} \sum_{x' \in \mathcal{X}_{r(v, \ell_1)}^1} x'^i |1_c, i, \ell_1\rangle \right) |0^{(n+5)}\rangle + |v\rangle |\perp\rangle |0\rangle \\
& \xrightarrow{I_{n-s} \otimes H_{t_1}^{\otimes s}} \frac{1}{2^{3/2} s} |v\rangle \sum_{i \in [4]} \left(\sum_{x' \in \mathcal{X}_{r(v, \ell_1)}^1} x'^i |0_c\rangle + \frac{1}{s} \sum_{x'' \in \mathcal{X}_v^2} x''^i |1_c\rangle \right) |i, 0^{(2n+5)}\rangle + |v\rangle |\perp\rangle |0\rangle \\
& \xrightarrow{(O_{K-1})_{v, k}} \frac{1}{2^{3/2} s} |v\rangle \sum_{i \in [4]} \left(\sum_{x' \in \mathcal{X}_{r(v, \ell_1)}^1} x'^i |0_c\rangle + \frac{1}{s} \sum_{x'' \in \mathcal{X}_v^2} x''^i |1_c\rangle \right) \frac{1}{k_v^c} |i, 0^{(2n+5)}\rangle + |v\rangle |\perp\rangle \\
& = |v\rangle \left(\sum_{i, c} \frac{1}{2^{3/2} s^c} \sum_{x_u \in \mathcal{X}_v^c} \frac{x_u^i}{k_v^c} |c, i, 0^{(2n+5)}\rangle + |\perp\rangle \right) \\
& = |v\rangle \left(\sum_{i, c} \frac{1}{2^{3/2} s^c} \varphi(\mathcal{X}_v^c) |c, i, 0^{(2n+5)}\rangle + |\perp\rangle \right) \\
& = |v\rangle \left(\sum_{j \in [8]} c_j m_{v, j} |j, 0^{(2n+5)}\rangle + |\perp\rangle \right) \quad \text{with } \mathbf{c} \text{ defined as in eq. (25).}
\end{aligned}$$

Complexity Analysis in Detail

We can construct a quantum circuit to implement oracles O_L and $\tilde{O}_X$ with $\mathcal{O}(\text{polylog } N + d')$ elementary gates [33].

TABLE II. Quantum Gate Complexity and Queries for Various Sources

Operator	Description	Gate Complexity (Count)	Qubit Count	Query Complexity
H	elementary gate	1	1	-
O_L	input model	$\mathcal{O}(\text{polylog } N + d')$	$\mathcal{O}(\log N)$	$\mathcal{O}(1)$
$\tilde{O}_X$	input model	$\mathcal{O}(\text{polylog } N + d')$	$\mathcal{O}(\log N + d')$	$\mathcal{O}(1)$
$\tilde{O}_K$	derived from O_L	$\mathcal{O}(\text{polylog } N + d')$	$\mathcal{O}(\log N + d')$	$\mathcal{O}(\log^2 s)$
F_X	type of R_y	$\mathcal{O}(d')$	$\mathcal{O}(d')$	-
$F_{K^{-1}}$	type of R_y (INV)	$\mathcal{O}(d')$	$\mathcal{O}(d')$	-
O_X	$1 \tilde{O}_X \cdot 5 \text{c-}F_X \cdot 1 \tilde{O}_X^T$	$\mathcal{O}(\text{polylog } N + d')$	$\mathcal{O}(\log N + d')$	-
$O_{K^{-1}}$	$1 \tilde{O}_K \cdot 1 F_{K^{-1}} \cdot 1 \tilde{O}_K^T$	$\mathcal{O}(\text{polylog } N + d')$	$\mathcal{O}(\log N + d')$	$\mathcal{O}(\log^2 s)$

The rotation operators are simply controlled versions of single-qubit $R_y(\theta)$ rotations, with the angle computed from the d' -qubits in the control register. Using simple fixed-point arithmetic, such operators scale as $\mathcal{O}(d')$.